

● ACTIVE



FlotBot

Endpoint Security Monitoring System

AI System Capability Report

Comprehensive overview of AI models, capabilities, and architecture

VERSION

2.0 — AI Upgraded

AI ENGINE

Dual-Provider (Cloud + Local)

PLATFORM

Electron (Linux / Windows)

DATE

August 2026

2

AI PROVIDERS

5

MONITOR
MODULES

20+

THREAT
RULES

100%

OFFLINE
CAPABLE

5s

SCAN CYCLE



What Is FlotBot AI?

FlotBot is an **AI-powered desktop security monitoring tool** that runs 24/7 on your machine. It combines two AI engines — one cloud (Gemini) and one local (Ollama) — to monitor your system in real-time, detect threats automatically, explain alerts in plain English, and guide you through security decisions. It works **even without internet** using local AI models.



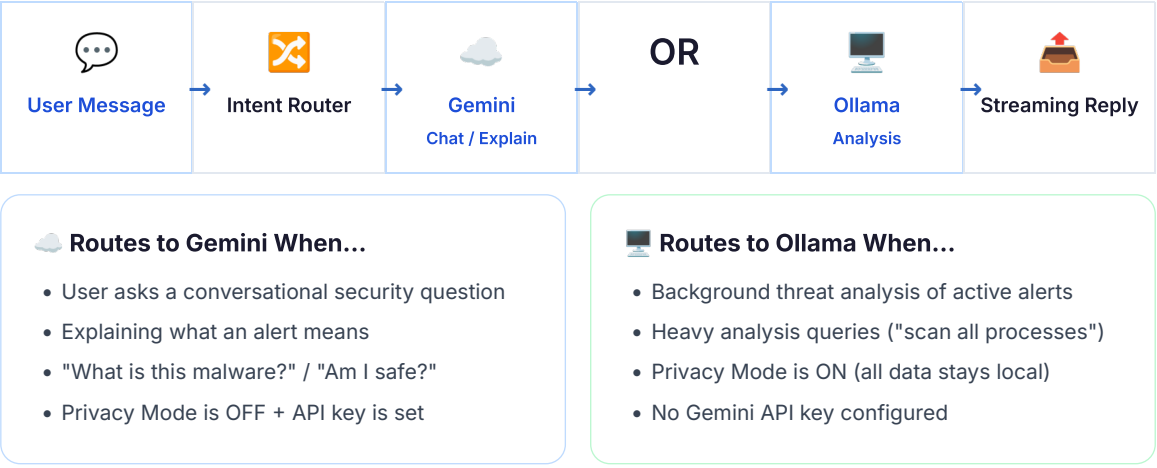
AI Models Used

MODEL	PROVIDER	WHERE IT RUNS	USED FOR	INTERNET?	STATUS
Gemini 2.0 Flash	Google DeepMind	☁️ Google Cloud	Chat conversations, alert explanations, security questions	✅ Required	Cloud AI

qwen2.5:3b / Mistral	Ollama (Local)	 Your Machine	Threat analysis, background scanning, privacy-mode chat	 Not needed	Local AI
MockProvider	Built-in FlotBot	 Your Machine	Offline keyword fallback when both providers are unavailable	 Not needed	Fallback

 **Smart Auto-Selection:** FlotBot automatically picks the best available model. If no Gemini API key is set, it silently switches to Ollama. If Ollama is offline, MockProvider takes over. The app never crashes regardless of provider state.

 How AI Routing Works (Smart Hybrid Engine)



✔ What The AI Can Do

🔍 Explain Security Alerts

When a threat is detected, click "🔍 Explain" and the AI explains it in plain English using everyday analogies — no technical jargon.

- 3-sentence beginner-friendly explanation
- Tells you exactly what to do
- Results cached for instant repeat access
- Streams word-by-word (ChatGPT-style)

💬 Security Chat Assistant

Ask FlotBot anything about your system security in natural language.

- "Is my system safe right now?"
- "What is ransomware?"
- "Check my running processes"
- "What apps are open?"
- Maintains conversation history (last 20 msgs)

📊 Threat Analysis Engine

Runs automatically every 5 seconds in the background. Uses Ollama to analyze all active alerts.

- Generates a 0–100 threat score
- Summarizes the threat landscape
- Provides top 3 recommended actions
- Only fires when alert state changes (efficient)

🛡️ Real-Time Alert Detection

Rule-based engine + AI augmentation detects threats before you notice them.

- Suspicious process execution (PowerShell, cmd)
- Network beaconing and C2 communication
- File system mass modification (ransomware)
- Persistence in autostart / cron / services

⚡ Streaming Responses

AI responses appear word-by-word in real-time — no waiting for the full reply.

- Tokens pushed via IPC events
- Blinking cursor shows active generation
- Abort old request when new one sent
- Cache replays instantly for repeat queries

🔒 Emergency Actions (UI Buttons)

High-impact security actions are performed via dedicated buttons — never via chat text (for safety).

- 🚫 **Terminate Process** by PID
- 🛑 **Block IP** via firewall
- ✔️ **Dismiss Alert** after review
- ⚙️ **Admin Console** (whitelist / privacy toggle)

📡 What FlotBot Monitors (5 Modules)

MODULE	WHAT IT WATCHES	LINUX COMMANDS USED	EXAMPLE THREATS CAUGHT
🖥️ System / Process	All running processes — PID, CPU, memory, command line, parent process	ps -eo pid,ppid,pcpu,pmem,rss,comm,args	Cryptominer, reverse shell, suspicious PowerShell

 Network	All active TCP/UDP connections — remote IPs, ports, states	<code>ss -tunp / /proc/net/tcp</code>	C2 beaconing, data exfiltration, port scanning
 File System	Downloads, Desktop, Documents, /tmp — file creation/modification	<code>Node.js fs.watch + fs.statSync</code>	Ransomware mass-encryption, executable drops
 Registry / Persistence	Linux: XDG autostart, systemd services, cron, shell RC files	<code>~/.config/autostart, /etc/systemd, crontab</code>	Backdoor persistence, malicious autostart entries
 Malware Correlation	Cross-module behavior analysis — correlates process + network + file events	Internal rule engine (no external tool)	APT patterns, multi-stage attacks, lateral movement

Privacy Modes & AI Behaviour

Online Mode (Default)

- **Chat:** Gemini 2.0 Flash (cloud)
- **Analysis:** Ollama (local, always)
- Requires: GEMINI_API_KEY in .env
- Best response quality for chat
- Threat data never leaves machine

Privacy Mode (Admin Toggle)

- **Chat:** Ollama (fully local)
- **Analysis:** Ollama (fully local)
- Zero cloud data — 100% offline
- Toggle from Admin Console → no restart
- Setting persists across app restarts

Auto-Fallback Mode

- No Gemini key → auto-uses Ollama for chat
- Ollama offline → uses MockProvider
- App never crashes, always responds
- Status badge shows current mode in UI
- Latency pill shows response time (ms)

AI Performance Improvements (v2.0)

Streaming tokens: First word appears in ~300ms — feels instant vs waiting 3–8s for full reply

Response caching: Repeated alert explanations served instantly from 2-min cache

Abort control: Sending a new message cancels the previous in-flight request immediately

Warm-up ping: Ollama model pre-loaded on startup — first real query has no cold-start delay

Parallel init: Both providers initialize simultaneously — startup is faster

Smart debounce: Background threat analysis only fires when alert state actually changes

Auto model selection: If configured model not installed, picks best available (qwen2.5, llama3, etc.)

Adaptive timeouts: Short chat = 12s timeout · Heavy analysis = 40s timeout

Quick Setup Guide

Step 1 — Prerequisites

- Node.js v18+ installed
- Ollama installed (ollama.com)
- Run: `ollama pull mistral`
- Or use existing: `qwen2.5:3b`

Step 2 — Configure

- Copy `.env.example` → `.env`
- Add `GEMINI_API_KEY=...` (optional)
- Get key free at `aistudio.google.com`
- Without key: Ollama handles everything

Step 3 — Run

- Open terminal in project folder
- Run: `npm start`
- Floating shield bubble appears
- Click bubble → chat panel opens